Forward modelling of dark matter density

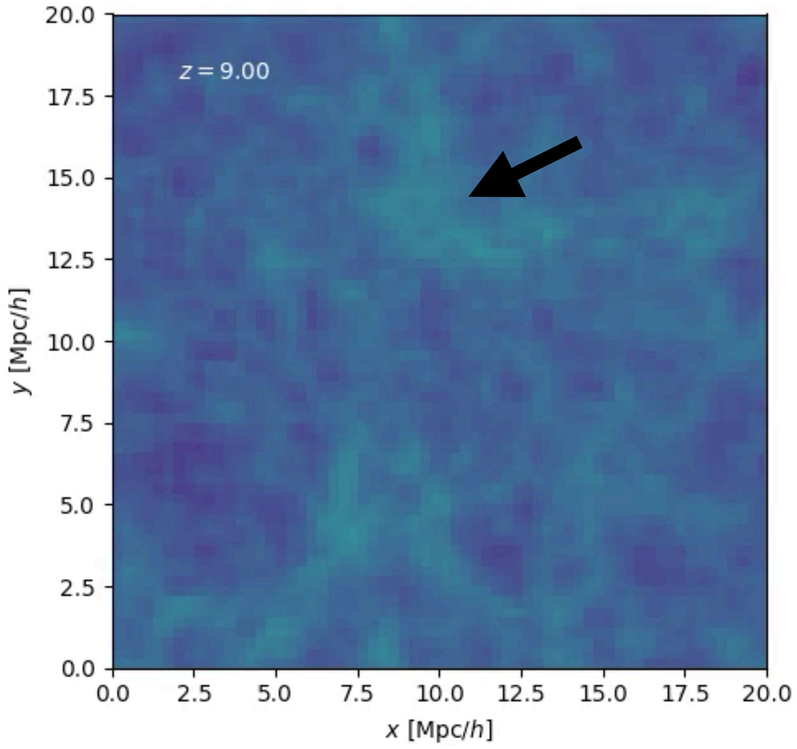
Reconstructing the accretion history of galaxy clusters from late-time observations

Forward model

Observations : Galaxy catalogues → galaxy 3D density maps

Aim : Infer current and past *dark matter* density map from current *galaxy* density map

Method : Using `JAX / Tensorflow`, optimise differentiable N-body simulations (gravitation only ≡ dark matter)



Starting from a realisation of a Gaussian Random Field, we run a N-body simulation that simulates the gravitational collapse of matter in the Universe through cosmic time.

Here on the left is represented a projection of the dark matter density. We can see cosmic structures forming through cosmic time.

The black arrow shows a massive galaxy cluster.

Thanks to GPU acceleration, such a simulation runs in ~0.2 s and is differentiable (10x speed-up, Modi+20)